

JY-CRPG-BENCH: LONG-HORIZON AGENCY IN AN OUT-OF-DISTRIBUTION OPEN-WORLD RPG

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ABSTRACT

Long-horizon agency requires a model to perceive an unfamiliar world, decide for itself what is worth doing, and hold one plan across hours of dependent decisions. We present JY-CRPG-BENCH, a benchmark that places a vision-language model inside 金庸群俠傳 (*Heroes of Jin Yong*, Heluo Studio 1996), an unmodified commercial open-world role-playing game under DOS emulation, and scores it against the ending the game defines, which requires recovering the fourteen books of the Jin Yong canon and takes a first-time human player tens of hours. The environment lies far outside the training distribution of current vision encoders, with a 320×200 framebuffer, a 45° isometric projection in which no key moves straight up the screen, Traditional Chinese rendered as 16-pixel bitmap glyphs, and objectives stated only inside its fiction. Every model plays through the same brief and control surface, with the frame as its observation and keys as its actions, and with no memory scaffold and no access to game state, so a run measures the model. Progress is read from the memory of the emulated machine and from the save the game writes for itself, so every milestone is an event that pixel similarity cannot manufacture, and one instrument spans the whole distance: a ladder of machine-verified milestones while models learn to move, and completion time once they finish. We report 8 frontier models against a seeded random baseline under a 60-minute budget, and release JY-CRPG-BENCH as a live, replayable public arena at <https://github.com/hanxiao/jy-crpg-bench>.

1 INTRODUCTION

Frontier models prove competition mathematics and write working compilers, yet the tasks they complete reliably are still measured in hours of human work (Kwa et al., 2025). Long-horizon evaluation takes two forms. Episodic benchmarks set a goal and score its completion; continuing benchmarks give the model an environment with no terminal state and score what accumulates (Backlund & Petersson, 2025; Fan et al., 2026). Open-world games belong to the second family and add a requirement the business simulations do not: the model has to recover its own state from raw perception, because nothing in the environment reports it.

Existing game benchmarks each remove one part of that problem to isolate another. BALROG evaluates six research environments in language-only and language-vision modes (Paglieri et al., 2025); TextQuests removes perception entirely (Phan et al., 2025); ARC-AGI-3 excludes language and cultural priors by construction (ARC Prize Foundation, 2026); VideoGameBench detects progress in real-time action games by matching against checkpoint images taken from walkthroughs (Zhang et al., 2025); MineExplorer filters out tasks whose solutions depend on game-specific knowledge (Ju et al., 2026). Each removal is principled, and each leaves the combination untested. We keep the parts together, because acting through unfamiliar perception, unfamiliar language and unfamiliar culture at once is what a deployed model is asked to do.

JY-CRPG-BENCH runs a single deep environment, following the precedent of Crafter and NetHack (Hafner, 2021; Küttler et al., 2020). The environment is 金庸群俠傳, a 1996 Chinese role-playing game that remains a landmark of its genre, executed as its original DOS binary under emulation with no modification. The game presents an open world built from the fourteen novels of Jin Yong: a two-tier map, several hundred named characters each carrying level, experience, skills, items and reputation, and an ending that requires recovering fourteen books scattered across the world. A first



Figure 1: The environment at the resolution the model receives it, upscaled three times without interpolation: the two tiers of the world (a, b), the four axes the movement keys run along (c), the three kinds of text panel through which every objective, choice and statistic is stated (d–f), the bag (g), the outdoor menu with the party list it opens (h), and a turn of battle with the menu and record of the acting character (i).

human playthrough takes tens of hours. The model receives the framebuffer and a key vocabulary through one fixed, minimal harness, and nothing else, so what varies between runs is the model.

We measure against that ending. The fourteen books sit behind items, factions and martial arts that have to be acquired in order, in a world that reports nothing about itself, and the game records whether they have been recovered. A model that finished would have held one plan across tens of hours of self-directed play, which is the capability long-horizon evaluation exists to detect. The instrument therefore covers the whole distance: a ladder of machine-verified milestones orders the field while models are learning to move through the world, and completion time orders it once they can finish, so the benchmark gains a speedrun ranking at the point where a pass-or-fail score would saturate.

Three properties make the environment adversarial for current models, and Figure 1 shows the world they apply to. Perception is out of distribution on several axes at once. Downsampling costs vision-language models accuracy (Niu et al., 2025), degraded input costs them more (Usama et al., 2025), out-of-distribution images degrade recognition even of familiar categories (Lin et al., 2026), and text rendered as pixels is understood less well than the same text supplied as tokens (Liu et al., 2026), and precise spatial reading fails once primitives are small or adjacent (Rahmanzadehgervi et al., 2024; Wang et al., 2024); a 320×200 frame of 1996 bitmap art carrying 16-pixel Traditional Chinese glyphs sits at the hard end of all five. Control is projected. The world is drawn in a 45° isometric projection, so the four movement keys run along screen diagonals, and a model that reasons in screen coordinates walks in circles. Progress is self-directed. The game keeps no quest log, draws no objective marker and reports no score. The brief supplies the opening route, from the first room to the compass, and a manual of the controls; from the compass on, deciding where to go and what is worth doing is part of the problem. ARC-AGI-3 places the same goal-setting requirement at the centre of agency (ARC Prize Foundation, 2026).

Table 1: Interactive agent benchmarks. *Observation* is what the model receives before any upscaling, *self-directed* marks environments that supply no task list, score or objective marker during play, and *progress from* is what a progress claim is checked against.

Benchmark	Environment	Observation	Language	Self-directed	Progress from
BALROG	6 research games	text, render	English	no	env. reward
TextQuests	interactive fiction	text	English	yes	game state
ARC-AGI-3	abstract puzzles	64×64 grid	none	yes	env. score
VideoGameBench	1990s action games	native px	English	no	checkpoint images
GameWorld	34 browser games	browser	English	no	game state
MineExplorer	Minecraft	3D render	English	no	milestone rules
JY-CRPG-BENCH (ours)	one 1996 CRPG	320×200 px	Trad. Chinese	yes	memory, save file

Measuring such an environment from the screen is unreliable, and the field has begun to say so: GameWorld names heuristic verification as the obstacle to standardised game-agent evaluation and pairs each of its tasks with a state-verifiable metric (Ouyang et al., 2026). We take the same position and push it into the emulator. The benchmark reads progress out of the memory of the running machine, from the character array the game maintains, and out of the save file the game writes when asked, which is where the party and the world position are true. The model has no call that returns any of this, and nothing read from the screen enters a score.

We make three contributions.

1. We present JY-CRPG-BENCH, a long-horizon benchmark on an unmodified commercial open-world role-playing game whose difficulty comes from named and independently checkable properties of the environment: out-of-distribution low-resolution perception, isometric control, Traditional Chinese grounding, a keyboard action space of which a small subset does anything, and persistent character state that only coherent play advances.
2. We score progress in the terms the game keeps. Character records, the party and the fourteen books are read from emulator memory and from a save the game performs for itself, validated against the archives the release ships, and measurement never writes to the machine it observes. Every rung of the ladder is an event the model has to cause, and the terminal rung is the ending the game defines.
3. We evaluate 8 frontier models under the same harness and against a seeded random baseline under a wall-clock budget, and release the benchmark, its harness and every recorded run as a public, replayable arena.

Table 1 places JY-CRPG-BENCH among interactive agent benchmarks. Section 2 reviews game environments as agent benchmarks and the measurement problem that long horizons create, Section 3 details the environment, the interface, the session protocol and the measurement, Section 4 reads the current field for the headroom, resolution and validity of the instrument, and Section 5 concludes.

2 RELATED WORK

2.1 GAME ENVIRONMENTS AS AGENT BENCHMARKS

Games have been the standard setting for sequential decision-making since the Arcade Learning Environment (Bellemare et al., 2013), and language and vision-language agents have produced a second generation of game benchmarks. BALROG aggregates six reinforcement-learning environments under one protocol and reports a persistent gap between what a model knows about a game and what it does inside one (Paglieri et al., 2025). Imgame-Bench isolates how much of a game result belongs to the harness (Hu et al., 2026), and Orak spans twelve popular titles with configurable agentic modules (Park et al., 2026). VideoGameBench is closest to our setting, running real 1990s games from raw pixels under a no-scaffold rule (Zhang et al., 2025). GameWorld standardises the action interface across 34 browser games and pairs every task with a state-verifiable metric (Ouyang et al., 2026). MineExplorer evaluates open-world exploration in Minecraft and finds that models handle single-hop tasks but degrade once hidden prerequisites must be coordinated over a longer

trajectory (Ju et al., 2026), which is the structure of the opening sequence in our environment as well. The closest precedents to our setting are the public Pokémon runs. Claude 3.7 Sonnet reached three gym badges from screen pixels behind a memory scaffold, where an earlier model had not left the first house (Anthropic, 2025); Gemini 2.5 Pro finished Pokémon Blue in 813 hours behind a harness that translated the RAM of the game into text, and an ablation without vision performed about as well (Comanici et al., 2025); the PokeAgent Challenge turned the setting into a speedrunning track scored by completion time over milestones (Karten et al., 2026a). Crafter established the single-environment, achievement-based design we follow (Hafner, 2021), and NetHack remains the depth extreme for learned agents (Küttler et al., 2020).

JY-CRPG-BENCH differs from these in running one persistent open world with Traditional Chinese as its only text, in verifying progress from emulator memory and from the save file of the game, and in scoring against the completion condition the game defines, which gives the leaderboard a second scale in completion time.

2.2 LONG-HORIZON EVALUATION AND ITS MEASUREMENT

Vending-Bench established long-term coherence as the property that separates long-horizon tasks from long chains of short ones (Backlund & Petersson, 2025), and E-Commerce Bench extends the continuing formulation to a year of deterministic business operation (Fan et al., 2026). Both keep perception out of the loop. TextQuests shares our self-directed, long-horizon framing while removing vision (Phan et al., 2025), TextAtari stretches horizons to 10^5 steps over text renderings (Li et al., 2025), ARC-AGI-3 isolates fluid agency by excluding language and culture (ARC Prize Foundation, 2026), and AutumnBench probes world-model quality through environment-level queries (Warrier et al., 2025). UltraHorizon makes exploration under hidden rules the task itself (Luo et al., 2025), and Kapoor et al. (2026) observe that benchmark design has favoured tasks that are precisely specified, automatically graded and short, and call for open-world evaluation. Navigation from rendered images is a documented weakness: on mazes, one-shot navigation collapses for every model by ten cells a side (Jiang et al., 2026), and the failures are attributed to looping (Einarsson, 2025).

Measurement is where long horizons strain evaluation. The ABC checklist catalogues task-validity and outcome-validity failures across agentic benchmarks (Zhu et al., 2025), an audit of agent traces finds reward hacking in a majority of them on two suites (Shao et al., 2026), and a survey of the horizon gap finds that the field answers uninformative outcome signals by building denser step-level ones (Chen et al., 2026). State-based verification has precedent outside games, where τ -bench compares the final database state with an annotated goal (Yao et al., 2024). Our milestone ladder is an instance of the same answer, read from machine state so that dense signals cannot be mistaken for progress.

3 JY-CRPG-BENCH

3.1 ENVIRONMENT

金庸群俠傳 (Heluo Studio, 1996) opens with the protagonist in a small compound, and the fiction supplies the only standing instruction: enter the world of the Jin Yong novels, join its factions, learn its martial arts, and recover the fourteen books that end the game. The world is two-tier, with many local scenes such as buildings, sects and caves attached to one outdoor map of 480×480 tiles (scarsty & contributors, 2026). For each of 320 named characters the game maintains a record carrying level, experience, health, stamina, learned skills, carried items, reputation and aptitude. Progression is nonlinear and gated. Every scene carries an entry condition, and at the start all of them are closed except the home, five inns and the house of the hermit 南賢; events open the rest, and the opening sets their order. The compound holds a searchable chest, one speaking character and one exit tile. Beyond it, the path south leads to the hermit, whose long conversation opens the house of the first companion, and a cabinet beside him holds the compass, the only source of coordinates the game offers. The companion joins after a yes-or-no prompt at which the confirm key means no. Fights are turn-based on the scene grid. Each turn a character moves along the four axes and then strikes, uses an item, waits or rests. Skills draw on inner force, and the first scripted fight opens by draining it. An automatic mode hands the character to the AI of the game, which alternates striking and resting and loses to an opponent who heals. A party that wanders elsewhere first meets

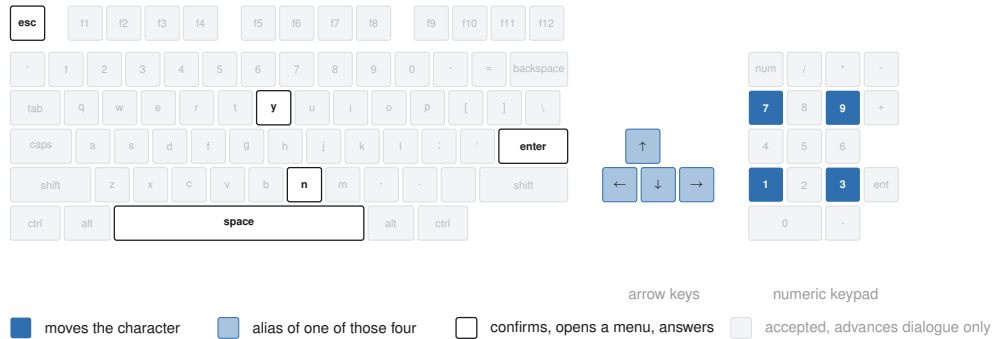


Figure 2: The action space. The API accepts 130 key names over 100 distinct keys; outside dialogue the game answers the nine picked out here, any key advances a dialogue, and a model has to find them.

storylines whose fights it cannot survive at level one, and a lost fight ends the game, which then offers nothing but its own load menu.

Figure 1 shows the world at the resolution the model receives it. Every objective, prompt and statistic reaches the model as Traditional Chinese rendered at sixteen pixels a line, and the character sheet it would have to read is the same record the benchmark reads from memory. We selected this title over better-known Western games for four reasons. Its text is Traditional Chinese throughout, a script that multimodal evaluation has largely passed over (Tam et al., 2025), so cross-lingual grounding is a first-class obstacle. It is menu-driven, so a model that deliberates between actions is not structurally disadvantaged as it would be in a real-time title (cf. Zhang et al., 2025). Its progression state is rich enough to measure at the level of individual character statistics. And its 45° isometric projection makes the mapping from screen to action something the model must infer.

3.2 OBSERVATION AND ACTION SPACE

The model observes the raw framebuffer at 320×200 and acts by pressing keys. A scored session serves the native frame and nothing else, so no information is added or smoothed away, and nothing is annotated: there is no set-of-marks overlay and no accessibility tree (cf. Xie et al., 2024). The action space is the DOS keyboard, 130 names over 100 distinct keys, and Figure 2 shows how few of them the game answers. Movement runs along the four numpad diagonals, with the arrow keys aliased onto the same axes, so a model that presses `up` expecting to walk up the screen moves up and to the right. Perception has traps of its own. The view follows the character except near the edge of a scene, where the background holds still and the sprite walks across it, so a model that reads its movement from the scrolling of the background loses count of the tiles it walked there. Canopies and roofs are drawn over the character, and the entrance of a building is a single tile on one face of its wall, so reaching a door means walking around the compound to the side the door is on.

There is one action. It presses a key, or a list of keys in order, holding each for a requested number of emulated frames, ten by default, and returns once the screen has reacted and held still, waiting through a scene transition if one starts, so the frame a model receives is the one to act on. A repeat is a list of the same key and a menu path is a list, and there is no call that lets the game run without input, because the game only moves on input and the action already waits for the result. The hold is counted in frames because the game samples its keyboard once per game-loop iteration and a shorter pulse can be consumed without registering. A scored session answers four HTTP calls, listed in Appendix A: the screen, the brief, the key names and the action. The development runner adds three snapshot calls that a scored session answers with 404, since a run with an out-of-band rewind is not a run. The save and load menu of the game stays open to the model, as to a human player, so a model that loses a fight can reload the last save the game wrote, a rewind within the run that the recording shows.

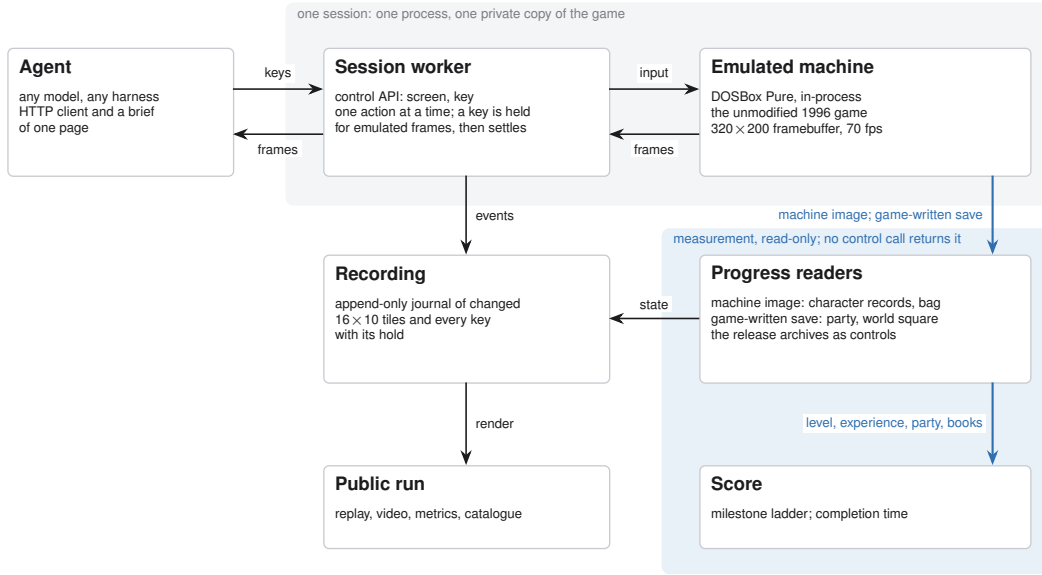


Figure 3: The apparatus. One process per session loads the game core in-process and answers the control API; the recording feeds the public replay, and progress is read from the emulated machine and from the save the game writes, on a path no control call reaches.

3.3 HARNESS AND SESSION PROTOCOL

The benchmark supplies the harness, and the harness is minimal. The brief is the whole instruction the model receives: the calls that constitute play, the rules of a run, the opening route as far as the compass, and a manual of the controls and systems of the game, served in English and Chinese with the on-screen Traditional terms of the game kept verbatim, so the language a model reasons in is not itself a handicap. The model reaches the game through two tools, which look at the screen and press one or more keys, and an action returns only its metadata, so the model looks when it decides to. There is no memory scaffold, no map, no pathfinder and no access to the state of the game. This separates the setting from the public Pokémon runs, where the harness translated RAM into text and did the perception (Comanici et al., 2025), and it is what makes the leaderboard a ranking of models: harness choice moves game results (Hu et al., 2026; Karten et al., 2026b), and on long-horizon tasks the variance a harness induces can exceed the variance between models (Zhang et al., 2026). A session is created under a declared name; the reference harness requires a thinking level, one of seven from off to max, at launch, and files model and level with the usage report of Section 3.5.

A session is created with a playtime, 60 minutes by default and up to twenty-four hours, and the clock starts at the first playable moment. Each session is a separate emulator process with a private copy of the game, which every host supplies itself as described in Appendix D, with its save slots reset to a new game, so no run inherits the progress of another. A session ends only when its budget lapses or it completes; a long pause between keys is the model’s to spend, and there is no inactivity cutoff. When the budget lapses the session renders its recording to video, computes its metrics, and appends itself to the public catalogue. Figure 3 shows the apparatus, and Appendix C lists the session parameters.

Because the control surface is plain HTTP, the harness is small and is released with the benchmark, and any client can drive a session for development or spectating. Every run publishes a time-compressed video and a keypress timeline, so the behaviour behind a score can be reviewed.

3.4 STATE-VERIFIABLE PROGRESS

The game keeps its persistent state in one archive: a base block with the world position, the party roster and the shared bag, followed by 320 character records of 182 bytes and an item table. We validated that layout against the opening machine image and the archives the release ships, where

Table 2: The milestone ladder. *Evidence* is the game state that credits the rung and *source* is where the benchmark reads it: the session record of the harness, the bag and character array in emulator memory, or the save the game writes on the world map.

Rung	Evidence	Source
<i>The opening, reached without a fight</i>		
acted	a key event was sent	session record
picked something up	any item count above its opening count	bag in memory
reached the world map	the game wrote its save	save
holds the compass	the compass in the bag	bag in memory
recruited a companion	a party roster longer than one	save
<i>The campaign, from the first fight on</i>		
gained experience	experience above zero	character record
reached level 2	level above one	character record
holds one of the fourteen	a book in the bag or carried by the protagonist	bag and record

only 2 of 320 records violate $hp \leq \max Hp \wedge mp \leq \max Mp$, and located the same array in the live emulated machine by anchoring on one character name and confirming the two neighbouring records at the correct stride, since the name recurs in the image and the layout moves between runs. The bag in front of that array is the working copy and moves the moment an item is taken, so the item count and the books held are current. Reading is passive: the benchmark serialises the machine but never loads state into it. The one thing it presses for itself is the save described next, which costs the game a few seconds of menu time outside the accounting of the run and leaves the screen as it found it.

For the party roster and the world position the benchmark reads the game’s own save, the archive it writes in the format already validated against the shipped ones. The game offers its save menu on the world map alone and announces this through the height of the menu, so an attempt opens the menu, counts its rows and withdraws where saving is not offered, leaving the screen unchanged; a save that lands is itself the evidence that the party has crossed onto the world map. The archive the game writes is decoded like the shipped ones, and neither it nor anything read from it is reachable through the control surface.

Progress is summarised as eight milestones read from this ground truth, in the spirit of the achievements of Crafter (Hafner, 2021), and Table 2 lists them with the evidence each rests on. Only the first concerns the harness; the rest are the numbers the game keeps about itself. The rungs fall into two horizons that the game itself draws. The opening needs no fight: the chest in the first room, the exit tile, the compass in the cabinet of the hermit, and the companion whose house that conversation opens and who joins through dialogue are reached by walking, reading and answering. The campaign begins with the first fight, which is what experience and levels require, and runs through the chains of items, factions and fights that the books sit behind. The default budget is sized to the opening and the extended budgets to the campaign. The rungs are listed in the order the opening imposes, but a run is credited with every rung it reaches, so the count does not depend on the path taken, and a rung whose instrumentation post-dates a run is reported as unmeasured, since an unmeasured zero and a true zero are different claims.

The ladder points at the ending of the game. The fourteen books are what the game requires before its final sequence can be played, and the archive records every book held, so the count of books carries the ladder from its last rung to the threshold of that sequence, and the fourteenth book is the completion event the benchmark times: the first read that finds all fourteen held latches the played time. Sessions run to twenty-four hours on the same protocol, and a session that completes carries that time, which orders a field in which every entrant passes. One instrument therefore covers both horizons, and the benchmark keeps measuring after the first model finishes.

That ladder is the frozen score of the short benchmark and it stays fixed. For the multi-hour budgets the protocol allows, the release adds a second, additive report over the same ground truth: a finer trajectory that records the scenes a run reached, the battles it survived, the skills it learned and the books it gathered, out to the ending, with the walkthrough’s hard prerequisites as dependency gates between them. Each component is scored against a fixed denominator with its coverage reported

beside it, so a run instrumented on only part of the trajectory cannot post a falsely high number, and a milestone a run never measured is marked unmeasured rather than counted as failed. The report never rescores the short ladder and never feeds back into the short field’s intervals; it exists so that a long run holding no book still leaves an ordered record of how far its plan carried. Its checkpoint contract and component weights are released with the benchmark.

3.5 EFFORT

The ladder is the score, and nothing read from the screen enters it. Beside it the session records effort. The wall clock of every decision call is kept, so the played time and the think time between calls are measured on the server for any client. The tokens a run spends are a property of the client, so the reference harness meters them: after a run it sums the usage the provider reported for every assistant turn on the active branch of its session, as input, output and cache tokens with a cost at the rates the operator declared, and files the total once with the model, thinking level, harness version and brief language, under the name the session was created with. A run with any unmetered turn files nothing, so the record carries the whole meter or none. Effort is shown beside a run and never ranks it.

A seeded random policy plays the same budget through the same control surface, drawing from thirteen keys the brief teaches, weighted towards movement and confirmation, and never reading the screen, following the convention that anchors a score on random and human play (Mnih et al., 2015; Hafner, 2021; ARC Prize Foundation, 2026). The rungs it reaches are what the environment gives away, and they are the floor of the ladder.

4 EVALUATION

The field at submission is 8 models, one session each at the 60-minute budget, and a seeded random baseline. A run is listed under the name declared when its session was created; two were created under names that were not the model, and are listed under the model the operator identified, Claude-3.7-Sonnet as `gemini-3.7-flash` and `pi-composer` as `grok-4.5`, with the declared names kept in the snapshot. No run in this field carries a usage report, so tokens and thinking level are not reported, and 2 of the 8 sessions fetched the brief from the session, in English, the rest reading the published page. Every session is public on the leaderboard with its video, replay timeline and metric record, and the catalogue snapshot the paper is generated from is committed beside the paper source. Appendix F lists the runs by name. This section reads the field for what it says about the instrument: how much of the ladder is open, how far it separates models, and how a budget is spent.

[The present field is one run per model. A sweep of repeated runs per model under the fixed harness, and a human reference run under the same budget and interface reported to the human-baseline checklist (Wei et al., 2025), are the two additions the field most needs.]

4.1 HEADROOM

Every rung is measured for the first time, and the world map is the highest the field reaches at this budget. 5 of 8 sessions pick something up, and 5 reach the world map, every one of them credited by the save the game wrote; 3 spend the whole budget in the scene they started in. No session holds the compass, recruits a companion, gains experience or holds a book: all 8 character records stand at level 1 with 0 experience and 1 skill, the state the session booted into, so every rung from the compass on is open. Figure 4 shows the ladder by model. Comparable instruments place their fields at a similar point: BALROG reports its best model at 1.5% progression on NetHack (Paglieri et al., 2025) and VideoGameBench reports the best vision-language models completing 0.48% of its suite (Zhang et al., 2025). The distance from the world map to the fourteen books is the headroom the benchmark holds.

4.2 DISCRIMINATION

Two rungs divide the field, and they divide it differently. The chest and the exit are both in the first room, and the sessions that searched one are not the sessions that found the other: 2 crossed without

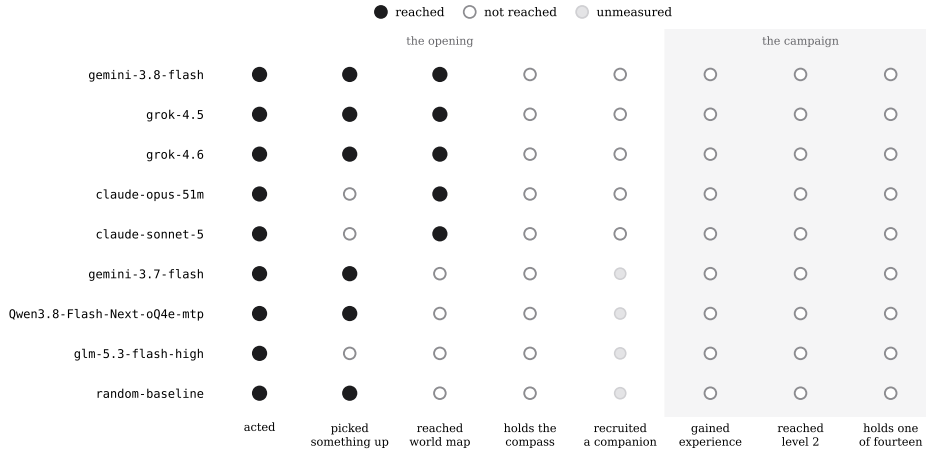


Figure 4: The milestone ladder, one row per model. Filled markers are rungs reached, hollow markers rungs not reached, and grey markers rungs the run carries no reading for.

the chest, 2 searched it without crossing, and 3 did both. The count of rungs is therefore a summary of what a run achieved and not a position on one path, which is why every rung reached is credited whatever the order. The random policy reaches 2 of the eight rungs in 60 minutes: it acts, and in a small room a random walk searches the chest within the hour. It never writes a save, so the world map is the first rung a random walk does not reach, and the chest rung separates models from one another but not from the floor. Above them the field is undivided: no session reaches the compass, which is where the ladder will next separate models.

4.3 USE OF THE BUDGET

The hour is enough for the first room and not for the compass. 5 sessions reached the world map and none reached the cabinet a short walk south of the exit; every session used its full budget. The record now carries the time of the first save the game writes on the world map, so the crossing time of later fields will be read from the game and not from the screen; the present field predates it. The protocol supports budgets to twenty-four hours, the setting in which the later rungs and a completion time come within reach, and deliberation is priced in wall clock, an efficiency framing shared with ARC-AGI-3 and TextQuests (ARC Prize Foundation, 2026; Phan et al., 2025).

[Long runs that exercise the additive long-horizon report are the measurement that separates a slow model from a stuck one; the protocol and the report already support them, and the field does not yet contain them.]

5 CONCLUSION

We presented JY-CRPG-BENCH, a long-horizon benchmark that runs an unmodified 1996 open-world role-playing game, gives every model the same brief and control surface, and scores it in the terms the game keeps about itself. The environment is adversarial in ways that are named and checkable: an out-of-distribution 320×200 isometric frame, Traditional Chinese as the only channel through which objectives are stated, and a world that supplies no score. Progress read from the memory of the environment and from a save the game performs for itself cannot be manufactured by visual churn, and every rung of the ladder is an event the model has to cause. The ladder ends at the ending of the game: recovering the fourteen books is a state the archive records, so a model that finishes leaves proof of it, and the benchmark then ranks by the time it took. Both scales exist from the start, so the first finishing model changes what the leaderboard measures, and the run that gets there is on record with its video and every key it pressed.

REFERENCES

Anthropic. Claude’s extended thinking. Research post, section “Claude plays Pokémon”, <https://www.anthropic.com/news/claude-extended-thinking>

- [//www.anthropic.com/research/visible-extended-thinking](https://www.anthropic.com/research/visible-extended-thinking), 2025.
- ARC Prize Foundation. ARC-AGI-3: A new challenge for frontier agentic intelligence. *arXiv preprint arXiv:2603.24621*, 2026.
- Axel Backlund and Lukas Petersson. Vending-Bench: A benchmark for long-term coherence of autonomous agents, 2025. URL <https://arxiv.org/abs/2502.15840>.
- Bahamut forum contributors. 原版金庸群俠傳攻略 (a complete walkthrough of the original game). <https://forum.gamer.com.tw/C.php?bsn=10859&snA=2778>, 2023.
- Marc G Bellemare, Yavar Naddaf, Joel Veness, and Michael Bowling. The arcade learning environment: An evaluation platform for general agents. *Journal of artificial intelligence research*, 47: 253–279, 2013.
- Mingguang Chen, Licheng Wang, and Bo Qu. The horizon gap: Planning, memory, execution, training, and evaluation for long-horizon LLM agents, 2026. URL <https://arxiv.org/abs/2608.06663>.
- Gheorghe Comanici, Eric Bieber, Mike Schaekermann, Ice Pasupat, Noveen Sachdeva, et al. Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities, 2025. URL <https://arxiv.org/abs/2507.06261>.
- Hafsteinn Einarsson. MazeEval: A benchmark for testing sequential decision-making in language models, 2025. URL <https://arxiv.org/abs/2507.20395>.
- Wei Fan, Xinjie Shen, Xudong Guo, Jianhong Tu, Yang Su, Yinger Zhang, Lianghao Deng, Fengyu Wang, Baohua Dong, Yangqiu Song, and Dayiheng Liu. E-Commerce Bench: Evaluating LLM agents on long-horizon autonomous business operation, 2026. URL <https://arxiv.org/abs/2608.30730>.
- Danijar Hafner. Benchmarking the spectrum of agent capabilities. *arXiv preprint arXiv:2109.06780*, 2021.
- Lanxiang Hu, Mingjia Huo, Yuxuan Zhang, Haoyang Yu, Eric P Xing, Ion Stoica, Tajana Rosing, Haojian Jin, and Hao Zhang. Igame-Bench: How good are LLMs at playing games? In *International Conference on Learning Representations*, volume 2026, pp. 81308–81356, 2026.
- Yuhan Jiang, Peng Luo, and Liqiu Meng. Lost in aggregation: A multi-scale diagnostic benchmark for LLM spatial navigation, 2026. URL <https://arxiv.org/abs/2606.22219>.
- Tianjie Ju, Yueqing Sun, Zheng Wu, Wei Zhang, Yaqi Huo, Xi Su, Qi Gu, Xunliang Cai, Gongshen Liu, and Zhuosheng Zhang. MineExplorer: Evaluating open-world exploration of MLLM agents in Minecraft, 2026. URL <https://arxiv.org/abs/2605.30931>.
- Sayash Kapoor, Peter Kirgis, Andrew Schwartz, Stephan Rabanser, J. J. Allaire, Rishi Bommasani, Harry Coppock, Magda Dubois, Gillian K Hadfield, Andrew B. Hall, Sara Hooker, Seth Lazar, Steve Newman, Dimitris Papailiopoulos, Shoshannah Tekofsky, Helen Toner, Cozmin Ududec, and Arvind Narayanan. Open-world evaluations for measuring frontier AI capabilities, 2026. URL <https://arxiv.org/abs/2605.20520>.
- Seth Karten, Jake Grigsby, Tersoo Upaa Jr, Junik Bae, Seonghun Hong, Hyunyoung Jeong, Jaeyoon Jung, Kun Kerdthaisong, Gyungbo Kim, Hyeokgi Kim, Yujin Kim, Eunju Kwon, Dongyu Liu, Patrick Mariglia, Sangyeon Park, Benedikt Schink, Xianwei Shi, Anthony Sistilli, Joseph Twin, Arian Urdu, Matin Urdu, Qiao Wang, Ling Wu, Wenli Zhang, Kunsheng Zhou, Stephanie Milani, Kiran Vodrahalli, Amy Zhang, Fei Fang, Yuke Zhu, and Chi Jin. The PokeAgent challenge: Competitive and long-context learning at scale, 2026a. URL <https://arxiv.org/abs/2603.15563>. NeurIPS 2025 Competition Track.
- Seth Karten, Joel Zhang, Tersoo Upaa Jr, Ruirong Feng, Wenzhe Li, Chengshuai Shi, Chi Jin, and Kiran Vodrahalli. Continual harness: Online adaptation for self-improving foundation agents, 2026b. URL <https://arxiv.org/abs/2605.09998>.

- Heinrich Küttler, Nantas Nardelli, Alexander Miller, Roberta Raileanu, Marco Selvatici, Edward Grefenstette, and Tim Rocktäschel. The NetHack learning environment. *Advances in Neural Information Processing Systems*, 33:7671–7684, 2020.
- Thomas Kwa, Ben West, Joel Becker, Amy Deng, Katharyn Garcia, Max Hasin, Sami Jawhar, Megan Kinniment, Nate Rush, Sydney Von Arx, Ryan Bloom, Thomas Broadley, Haoxing Du, Brian Goodrich, Nikola Jurkovic, Luke Harold Miles, Seraphina Nix, Tao Lin, Chris Painter, Neev Parikh, David Rein, Lucas Jun Koba Sato, Hjalmar Wijk, Daniel M. Ziegler, Elizabeth Barnes, and Lawrence Chan. Measuring AI ability to complete long software tasks, 2025. URL <https://arxiv.org/abs/2503.14499>. NeurIPS 2025.
- Wenhao Li, Wenwu Li, Chuyun Shen, Junjie Sheng, Zixiao Huang, Di Wu, Yun Hua, Wei Yin, Xiangfeng Wang, Hongyuan Zha, et al. TextAtari: 100k frames game playing with language agents. *arXiv preprint arXiv:2506.04098*, 2025.
- Ling Lin, Yang Bai, Heng Su, Congcong Zhu, Yaoxing Wang, Yang Zhou, Huazhu Fu, and Jingrun Chen. OODBench: Out-of-distribution benchmark for large vision-language models, 2026.
- Qing’an Liu, Juntong Feng, Yuhao Wang, Xinzhe Han, Yujie Cheng, Yue Zhu, Haiwen Diao, Yunzhi Zhuge, and Huchuan Lu. VISTA-Bench: Do vision-language models really understand visualized text as well as pure text?, 2026.
- Haotian Luo, Huaisong Zhang, Xuelin Zhang, Haoyu Wang, Zeyu Qin, Wenjie Lu, Guozheng Ma, Haiying He, Yingsha Xie, Qiyang Zhou, Zixuan Hu, Hongze Mi, Yibo Wang, Naiqiang Tan, Hong Chen, Yi R. Fung, Chun Yuan, and Li Shen. UltraHorizon: Benchmarking agent capabilities in ultra long-horizon scenarios, 2025. URL <https://arxiv.org/abs/2509.21766>.
- Volodymyr Mnih, Koray Kavukcuoglu, David Silver, Andrei A. Rusu, Joel Veness, Marc G. Belle-mare, Alex Graves, Martin Riedmiller, Andreas K. Fidjeland, Georg Ostrovski, Stig Petersen, Charles Beattie, Amir Sadik, Ioannis Antonoglou, Helen King, Dharshan Kumaran, Daan Wierstra, Shane Legg, and Demis Hassabis. Human-level control through deep reinforcement learning. *Nature*, 518(7540):529–533, 2015.
- Junbo Niu, Yuanhong Zheng, Ziyang Miao, Hejun Dong, Chunjiang Ge, Hao Liang, Ma Lu, Bohan Zeng, Qiahao Zheng, Conghui He, and Wentao Zhang. Native visual understanding: Resolving resolution dilemmas in vision-language models, 2025.
- Mingyu Ouyang, Siyuan Hu, Kevin Qinghong Lin, Hwee Tou Ng, and Mike Zheng Shou. Game-World: Towards standardized and verifiable evaluation of multimodal game agents, 2026. URL <https://arxiv.org/abs/2604.07429>.
- Davide Paglieri, Bartłomiej Cupiał, Samuel Coward, Ulyana Piterbarg, Maciej Wołczyk, Akbir Khan, Eduardo Pignatelli, Łukasz Kuciński, Lerrel Pinto, Rob Fergus, et al. BALROG: Benchmarking agentic LLM and VLM reasoning on games. In *International Conference on Learning Representations*, volume 2025, pp. 96666–96702, 2025.
- Dongmin Park, Minkyu Kim, Beongjun Choi, Junhyuck Kim, Keon Lee, Jonghyun Lee, Inkyu Park, ByeongUk Lee, Jaeyoung Hwang, Jaewoo Ahn, et al. Orak: A foundational benchmark for training and evaluating LLM agents on diverse video games. In *International Conference on Learning Representations*, volume 2026, pp. 60684–60744, 2026.
- Long Phan, Mantas Mazeika, Andy Zou, and Dan Hendrycks. TextQuests: How good are LLMs at text-based video games? *arXiv preprint arXiv:2507.23701*, 2025.
- Pooyan Rahmazadehgervi, Logan Bolton, Mohammad Reza Taesiri, and Anh Totti Nguyen. Vision language models are blind: Failing to translate detailed visual features into words, 2024. URL <https://arxiv.org/abs/2407.06581>.
- scarsty and contributors. kys-cpp: a C++ reimplement of 金庸群俠傳. GitHub repository, <https://github.com/scarsty/kys-cpp>, 2026.
- Jiaqi Shao, Hanck Chen, Wei Zhang, Maxm Pan, and Bing Luo. Do agent benchmarks measure capability? Protocol validity in the age of agentic AI, 2026. URL <https://arxiv.org/abs/2607.22368>.

- Zhi Rui Tam, Ya-Ting Pai, Yen-Wei Lee, and Yun-Nung Chen. VisTW: Benchmarking vision-language models for Traditional Chinese in Taiwan, 2025. URL <https://arxiv.org/abs/2503.10427>.
- Muhammad Usama, Syeda Aishah Asim, Syed Bilal Ali, Syed Talal Wasim, and Umair Bin Mansoor. Analysing the robustness of vision-language models to common corruptions, 2025.
- Jiayu Wang, Yifei Ming, Zhenmei Shi, Vibhav Vineet, Xin Wang, Yixuan Li, and Neel Joshi. Is a picture worth a thousand words? Delving into spatial reasoning for vision language models, 2024. URL <https://arxiv.org/abs/2406.14852>. NeurIPS 2024.
- Archana Warriar, Dat Nguyen, Michelangelo Naim, Moksh Jain, Yichao Liang, Karen Schroeder, Cambridge Yang, Joshua B Tenenbaum, Sebastian Vollmer, Kevin Ellis, et al. Benchmarking world-model learning with environment-level queries. *arXiv preprint arXiv:2510.19788*, 2025.
- Kevin L. Wei, Patricia Paskov, Sunishchal Dev, Michael J. Byun, Anka Reuel, Xavier Roberts-Gaal, Rachel Calcott, Evie Coxon, and Chinmay Deshpande. Recommendations and reporting checklist for rigorous & transparent human baselines in model evaluations, 2025. URL <https://arxiv.org/abs/2506.13776>. ICML 2025.
- Tianbao Xie, Danyang Zhang, Jixuan Chen, Xiaochuan Li, Siheng Zhao, Ruisheng Cao, Toh J Hua, Zhoujun Cheng, Dongchan Shin, Fangyu Lei, et al. OSWorld: Benchmarking multimodal agents for open-ended tasks in real computer environments. *Advances in Neural Information Processing Systems*, 37:52040–52094, 2024.
- Shunyu Yao, Noah Shinn, Pedram Razavi, and Karthik Narasimhan. τ -bench: A benchmark for tool-agent-user interaction in real-world domains, 2024. URL <https://arxiv.org/abs/2406.12045>.
- Alex L Zhang, Thomas L Griffiths, Karthik R Narasimhan, and Ofir Press. VideoGameBench: Can vision-language models complete popular video games? *arXiv preprint arXiv:2505.18134*, 2025.
- Yunbei Zhang, Janet Wang, Yingqiang Ge, Weijie Xu, Jihun Hamm, and Chandan K. Reddy. Stop comparing LLM agents without disclosing the harness, 2026. URL <https://arxiv.org/abs/2605.23950>.
- Yuxuan Zhu, Tengjun Jin, Yada Pruksachatkun, Andy Zhang, Shu Liu, Sasha Cui, Sayash Kapoor, Shayne Longpre, Kevin Meng, Rebecca Weiss, et al. Establishing best practices in building rigorous agentic benchmarks. *Advances in Neural Information Processing Systems*, 38, 2025.

A THE BRIEF AND THE CONTROL SURFACE

A reference-harness run receives the tool adapter note and the text the game server serves at `/api/help`: the calls that constitute play, the rules of a run covering budget, recording and publication, the isometric key mapping with an explicit warning that arrow-key intuitions do not hold, the opening route as far as the compass, and a manual of the controls, menus, combat and attributes of the game. The page at `agents.md` is generated from the same source for clients that create their own sessions, and adds session creation and the note that any starting name is acceptable. The English and Chinese versions are equivalent in rules, with the on-screen Traditional terms preserved in both; the reference harness serves the Chinese brief by default, and the record of a run carries the language of every brief it fetched. Table 3 lists the control surface; a scored session answers the last three calls with 404, can read its remaining time and its own action counters, and is refused every progress field until the run ends. One more call, `POST /usage`, belongs to the broker that creates sessions rather than to a session: after a run, the harness files the usage meter of Section 3.5 under the name the session was created with, and the broker merges it into the catalogue entry of that run.

B RUNG DEFINITIONS AND THE SAVE PROCEDURE

A key is held for a requested 5 to 1200 emulated frames, 10 by default, with six frames between the keys of a list, and an action returns once the screen has reacted and held still, waiting up to 300

Table 3: The control surface. Seven calls, one way to do each thing, and one action; a scored session answers the first four.

Call	Body	Meaning
GET /api/screen	–	the current frame
GET /api/help	–	the brief, in English or Chinese
GET /api/keys	–	every accepted key name
POST /api/key	{"key": "kp3"}	press one key, with an optional hold
POST /api/key	{"key": ["kp9", "enter"]}	press several keys in order
GET /api/slots	–	emulator snapshots on disk
POST /api/save	{"name": "..."}	take a snapshot
POST /api/load	{"name": "..."}	restore a snapshot

frames for a black screen to be redrawn when a scene changes; nothing the server observes about the screen enters any score. The recording journal logs every key the server presses on the core with its hold, and the seeded sequence of a random run is recomputed offline and compared with that journal: for the random runs reported here the two agree key for key, and on the local runner single ten-frame taps of every call shape moved the character one tile per tap in the position the game saved. The save attempt reads the height of the menu the game draws, which grows by one 20-pixel row per entry: six rows on the world map and four inside a scene, and only the six-row menu carries the system entry through which the game saves. An attempt opens the menu and reads its height, withdrawing on the four rows of a scene and saving only where the six of the world map appear, so what admits a save is the game’s own menu and not a reading of where the party stands; and a look the model takes waits on the same single-player lock the attempt holds, so the model never observes the screen mid-save. Attempts wait for a two-second gap between the keys the model sends, recur every two minutes, and are made once more shortly before the budget lapses, into a save slot reserved for the benchmark, and the time of the first save is kept as the time of the crossing. The item rung compares the live bag with the bag at the first read of the run; the compass rung looks for item 182 in the bag, whose name in the shipped item table is 羅盤, and the book rung for items 144–157 in the bag or the four carried slots of the protagonist, each read from the live machine and corroborated by the save the game writes, and kept once seen; and the shipped names of all fifteen items are checked by the test suite.

C SESSION AND INFRASTRUCTURE PARAMETERS

The default budget is 3600 s, raised from 1200 s in September 2026, and is configurable to 24 h; there is no inactivity teardown; the concurrency cap is 24 sessions per 8-vCPU host, set from a measurement of 32 sessions holding native 70.09 fps on 8 GiB with memory binding before CPU, at about 123 MB of game copy per run, and the service runs on 16 GiB. Emulation is DOSBox Pure through libretro at a fixed 26800 cycles, the speed of a 486DX2-66 and period-correct for the game, with no sound card, so the observation is visual only, and with per-session private copies of the game and save directories. Recordings are stored as 16×10 -tile zlib deltas; published video is the native 320×200 frame above a 32-pixel strip naming the actor, the action number, the keys held and the clock, played at 8 times speed, or faster for long runs, with a timeline sidecar of a few kilobytes carrying every keypress and its hold duration for the scrubbable replay. A character-state read costs about 1.2 ms through serialisation into a reused buffer and is taken after every action, with gains latched against the opening bag, so an item picked up and later consumed still credits its rung. Outside scored runs the server can checkpoint and resume a session and serve many visitors from private workers behind one gateway; a scored run has neither. The leaderboard is a static page polling published JSON; only the live viewers of a running session, capped at twelve, reach its worker.

D RELEASE AND LICENSING

The harness code is released under MIT. The emulation core (`dosbox_pure_libretro`) is GPLv2, loaded at runtime through the libretro C API and redistributed unmodified; the bundled macOS core is built from `schellingb/dosbox-pure` at `7f6e8fb`, and the hosted service

downloads the libretto buildbot Linux build current at its image date. The game is the 1996 commercial release, copyright 智冠科技 and 河洛工作室. It is not redistributable, it is not tracked in the repository, and every host supplies its own copy. The public catalogue therefore carries our own measurement output, meaning metrics, videos and replay timelines, and nothing that substitutes for the game data. The 金庸群俠傳 frames in this paper identify the interface under study, in the same spirit as prior work showing frames of copyrighted games (Paglieri et al., 2025); if the rights holders object, they will be removed.

E THE FOURTEEN BOOKS

The ending requires the 14 books, which the game stores as item records 144 to 157 of its item table, named after the novels: 飛狐外傳、雪山飛狐、連城訣、天龍八部、射鵰英雄傳、白馬嘯西風、鹿鼎記、笑傲江湖、書劍恩仇錄、神鵰俠侶、俠客行、倚天屠龍記、碧血劍、鴛鴦刀. A book counts as held when it is in the shared bag or in the four carried slots of the protagonist, which is how the final rung of the ladder is read. With all fourteen in hand and sufficient reputation, the game opens its final sequence at a shrine on the far side of the map, where the books are placed and a last battle decides the ending (Bahamut forum contributors, 2023).

F THE FIELD AT SUBMISSION TIME

Table 4 lists every published run of the field, one row per model. The two runs created under a name that was not the model, Claude-3.7-Sonnet as gemini-3.7-flash and pi-composer as grok-4.5, are listed under the model, and the catalogue snapshot the paper is generated from, committed beside the paper source, keeps the declared names. The live leaderboard at <https://hanxiao.io/jy-crpg-bench/> shows the field and the runs published since.

An earlier field of 19 sessions at a 20-minute budget, 10 models under 13 labels, three of them carrying the prefix of the client that drove the control surface, predates the compass, companion, book and completion rungs and the usage report; its world-map verdicts rest on a screen heuristic and are not carried here, and 0 of its sessions gained experience. Its snapshot is committed beside the current one, and its media remain in the public bucket under an archive prefix.

Table 4: The field, one row per model. *played* is minutes of play, *ended* is how the session ended, *rungs* is the count of rungs reached, and the eight columns mark each rung: ✓ reached, ○ not reached, and – no reading. Rows are ordered by rungs reached and then by played time, with the random floor last.

Model	played	ended	rungs	acted	item	map	compass	party	exp	lv 2	book
gemini-3.8-flash	60	time	3/8	✓	✓	✓	○	○	○	○	○
grok-4.6	60	time	3/8	✓	✓	✓	○	○	○	○	○
grok-4.5	55	time	3/8	✓	✓	✓	○	○	○	○	○
gemini-3.7-flash	60	time	2/8	✓	✓	○	○	–	○	○	○
claude-opus-51m	60	time	2/8	✓	○	✓	○	○	○	○	○
Qwen3.8-Flash-Next-oQ4e-mtp	59	time	2/8	✓	✓	○	○	–	○	○	○
claude-sonnet-5	54	time	2/8	✓	○	✓	○	○	○	○	○
glm-5.3-flash-high	1	idle	1/8	✓	○	○	○	–	○	○	○
random-baseline	60	time	2/8	✓	✓	○	○	–	○	○	○